



DATA CLEANING AND PREPARATION PROJECT

DECODELABS DATA ANALYTICS INTERNSHIP

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TOOLS USED: PYTHON | PANDAS | NUMPY | MATPLOTLIB | JUPYTER NOTEBOOK

PROJECT OVERVIEW

Objective:

The goal of this project was to transform a raw transactional dataset into a clean and analysis-ready dataset.

Key Activities:

- Dataset inspection
- Missing value analysis
- Duplicate record validation
- Datatype verification
- Outlier analysis
- Final Dataset preparation

Outcome:

A high-quality cleaned dataset was generated for further analysis

DATASET UNDERSTANDING

Dataset Information:

- Customer transaction dataset
- Order information
- Product details
- Pricing data
- Shipping information
- Coupon usage details

Initial Assessment:

- Dataset structure reviewed
- Statistical summary generated
- Data quality issues identified.

```
<class 'pandas.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   OrderID               1200 non-null   str
1   Date                  1200 non-null   datetime64[us]
2   CustomerID            1200 non-null   str
3   Product               1200 non-null   str
4   Quantity              1200 non-null   int64
5   UnitPrice             1200 non-null   float64
6   ShippingAddress       1200 non-null   str
7   PaymentMethod         1200 non-null   str
8   OrderStatus           1200 non-null   str
9   TrackingNumber        1200 non-null   str
10  ItemsInCart           1200 non-null   int64
11  CouponCode            891 non-null    str
12  ReferralSource        1200 non-null   str
13  TotalPrice            1200 non-null   float64
dtypes: datetime64[us](1), float64(2), int64(2), str(9)
memory usage: 131.4 KB
```

	Date	Quantity	UnitPrice	ItemsInCart	TotalPrice
count	1200	1200.000000	1200.000000	1200.000000	1200.000000
mean	2024-03-22 16:58:48	2.945833	356.412750	5.485000	1053.968300
min	2023-01-01 00:00:00	1.000000	11.390000	1.000000	11.390000
25%	2023-08-03 18:00:00	2.000000	186.062500	4.000000	410.520000
50%	2024-03-23 00:00:00	3.000000	364.210000	5.000000	823.615000
75%	2024-11-08 12:00:00	4.000000	521.570000	7.000000	1578.475000
max	2025-06-30 00:00:00	5.000000	699.930000	10.000000	3456.400000
std	NaN	1.407557	197.177146	2.281983	819.856558

DATA CLEANING PROCESS

Cleaning Steps Performed:

1. Missing Value Analysis
 - Identified missing values in CouponCode
2. Missing Value Treatment
 - Replaced null values with "No Coupon"
3. Duplicate Record Check
 - Checked dataset for duplicate entries
4. Data Type Validation
 - Verified consistency of all variables.

```
df.isnull().sum()
```

✓ 0.0s

OrderID	0
Date	0
CustomerID	0
Product	0
Quantity	0
UnitPrice	0
ShippingAddress	0
PaymentMethod	0
OrderStatus	0
TrackingNumber	0
ItemsInCart	0
CouponCode	309
ReferralSource	0
TotalPrice	0
dtype:	int64

DATA QUALITY VALIDATION

Validation Activities:

- Duplicate analysis
- Data type verification
- Business rule verification
- Dataset consistency review

Key Findings:

- Dataset structure remained intact
- Missing values handled successfully
- No major quality issues observed

```
duplicates = df.duplicated().sum()  
  
print("Total Duplicate Rows: ", duplicates)
```

✓ 0.0s

Total Duplicate Rows: 0

df.dtypes

✓ 0.0s

```
OrderID          str  
Date             datetime64[us]  
CustomerID       str  
Product          str  
Quantity         int64  
UnitPrice        float64  
ShippingAddress  str  
PaymentMethod    str  
OrderStatus      str  
TrackingNumber   str  
ItemsInCart      int64  
CouponCode       str  
ReferralSource   str  
TotalPrice       float64  
dtype: object
```

OUTLIER ANALYSIS

Objective:

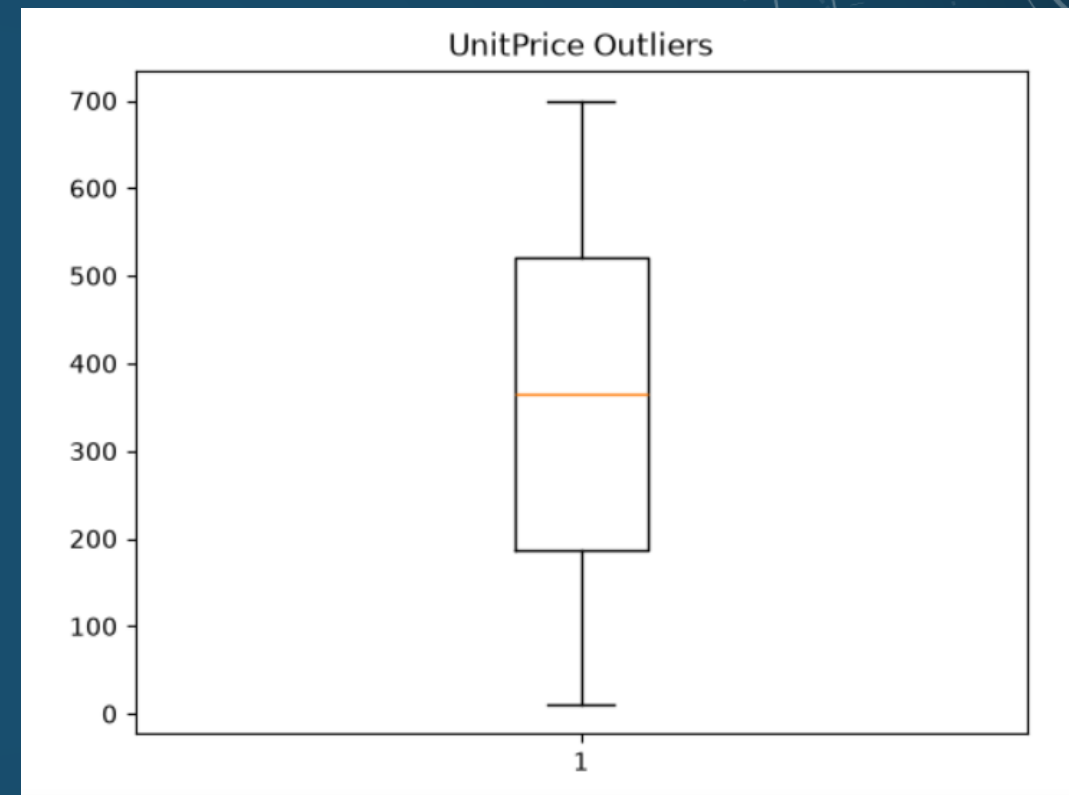
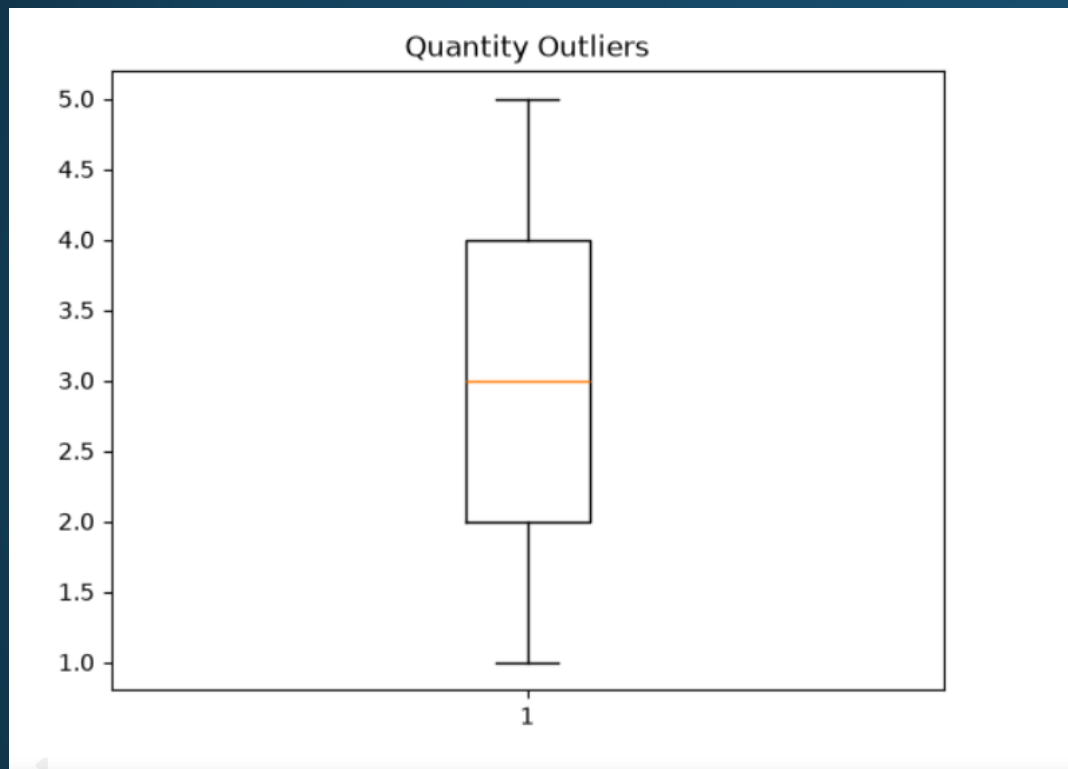
Identify unusual values that could affect data quality and future analysis.

Methods:

- Quantity boxplot
- Unit Price boxplot

Observations:

Boxplot analysis were performed for Quantity and Unit Price. No significant outliers were observed, indicating that the values fall within a reasonable business range and do not require treatment.



CONCLUSION

Project Summary:

- Dataset successfully cleaned
- Missing values treated
- Duplicates reviewed
- Data types validated
- Outliers analyzed
- Cleaned dataset exported

Final Outcome:

The dataset is now suitable for Exploratory Data Analysis (EDA), dashboard development and predictive analytics projects.

Thank You..